

Data Quality Report

Dataset: eBay Kleinanzeigen Used Cars | Generated: June 25, 2026 09:22

Executive Summary

371,528 Total Records	291,570 Issues Detected	27,577 Records Removed
343,951 Clean Records	92.6% Data Retained	14 Rules Checked

This report documents the findings of an automated data quality audit run on the eBay Kleinanzeigen Used Cars dataset. The pipeline profiled, validated, and cleaned 371,528 raw records across 20 columns, identifying 291,570 data quality issues across 14 validation rules. After cleaning, 343,951 records (92.6% of the original dataset) were retained as analysis-ready data.

Phase 1 - Data Profile

The raw dataset contained 371,528 rows across 20 columns. 4 exact duplicate rows were identified during profiling.

Missing Values by Column

Column	Missing Count	Missing %
vehicleType	37,869	10.19%
gearbox	20,209	5.44%
model	20,484	5.51%
fuelType	33,386	8.99%
notRepairedDamage	72,060	19.4%

Numeric Column Ranges

Column	Min	Max	Mean
price	0.0	2,147,483,647.0	17,295.1
yearOfRegistration	1,000.0	9,999.0	2,004.6
powerPS	0.0	20,000.0	115.5
kilometer	5,000.0	150,000.0	125,618.7
monthOfRegistration	0.0	12.0	5.7
nrOfPictures	0.0	0.0	0.0

postalCode	1,067.0	99,998.0	50,820.7
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Phase 2 - Validation Rules Engine

14 validation rules were run against the dataset. A total of 291,570 issues were detected: 3 HIGH severity, 6 MEDIUM severity, and 5 LOW severity.

* HIGH SEVERITY

Severity	Rule Description	Issues Found
HIGH	Exact duplicate rows	4
HIGH	Price outside valid range (EUR100 - EUR150,000)	13,552
HIGH	Year of registration outside valid range (1950-2016)	14,969

* MEDIUM SEVERITY

Severity	Rule Description	Issues Found
MEDIUM	Engine power outside valid range (10-1,000 PS)	41,346
MEDIUM	Missing vehicle type	37,869
MEDIUM	Missing fuel type	33,386
MEDIUM	Missing car model	20,484
MEDIUM	Missing damage status (notRepairedDamage)	72,060
MEDIUM	Listings that are 'wanted ads' not actual offers	12

* LOW SEVERITY

Severity	Rule Description	Issues Found
LOW	Month of registration outside valid range (1-12)	37,675
LOW	Missing gearbox type	20,209
LOW	nrOfPictures column is 0 for all rows - no information	1
LOW	Listings from commercial sellers (may skew price analysis)	3
LOW	Kilometer reading below 5,000 (suspiciously low for used car)	0

Phase 3 - Automated Cleaning

The cleaning pipeline applied 10 automated steps to resolve identified issues. 27,577 rows were removed from the dataset, retaining 343,951 clean records (92.6% of original data). The cleaned dataset was saved to output/clean/autos_clean.csv.

Cleaning Action	Records Affected
Duplicate rows removed	4
Invalid price rows removed	13,552
Invalid registration year rows removed	14,012
Invalid powerPS values nulled	32,524
Invalid month values nulled	28,457
Wanted ad listings removed	6
Commercial seller rows removed	3
Columns dropped (nrOfPictures)	1
vehicleType nulls filled with unknown	19,067
gearbox nulls filled with unknown	14,858
fuelType nulls filled with unknown	23,433
model nulls filled with unknown	15,801
notRepairedDamage filled with unknown	59,183
Text columns standardized to lowercase	6

Pipeline Result

The automated pipeline successfully processed 371,528 raw records, detected 291,570 data quality issues, and delivered 343,951 clean, analysis-ready records - retaining 92.6% of the original dataset with all critical quality issues resolved.

Recommendations

1. Implement upstream validation

Add price and year range constraints at the point of data entry to prevent corrupted records from entering the system in future.

2. Make damage status mandatory

The notRepairedDamage field was missing for 19.4% of listings. This field directly impacts pricing and buyer decisions - it should be required.

3. Standardize registration date capture

37,675 records had invalid month values (0 or >12). A dropdown selector at entry would eliminate this class of error entirely.

4. Schedule monthly pipeline runs

This pipeline is fully reusable. Running it monthly on new data ensures quality issues are caught early rather than accumulating over time.